



មហាវិទ្យាល័យវិស្វកម្ម
FACULTY OF ENGINEERING

Computer Programming

Lecture 6 Selections

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Introduction to Selection

- The program can decide which statements to execute based on a condition.

```
if (radius < 0)
{
    cout << "Incorrect input" << endl;
}
else
{
    area = radius * radius * PI;
    cout << "The area for the circle of radius " << radius
        << " is " << area << endl;
}
```

The bool Data Type

- The bool data type declares a variable with the value either **true** or **false**.
- Relational Operators

<i>Operator</i>	<i>Mathematics Symbol</i>	<i>Name</i>	<i>Example (radius is 5)</i>	<i>Result</i>
<	<	less than	<code>radius < 0</code>	<code>false</code>
<=	≤	less than or equal to	<code>radius <= 0</code>	<code>false</code>
>	>	greater than	<code>radius > 0</code>	<code>true</code>
>=	≥	greater than or equal to	<code>radius >= 0</code>	<code>true</code>
==	=	equal to	<code>radius == 0</code>	<code>false</code>
!=	≠	not equal to	<code>radius != 0</code>	<code>true</code>

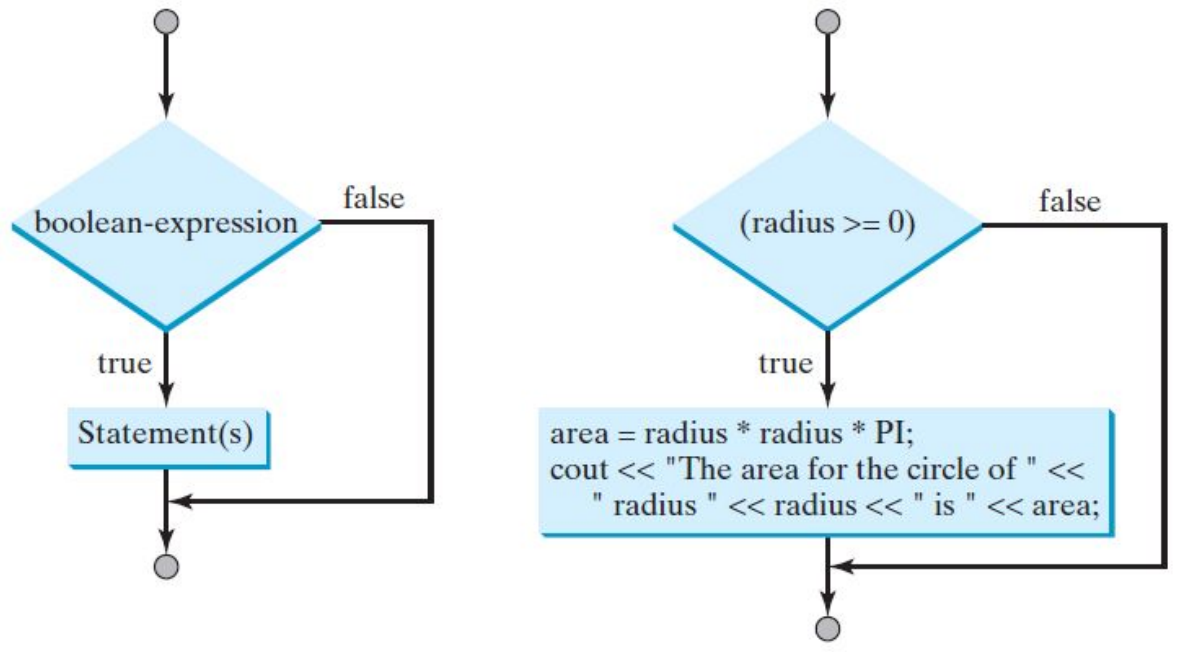
if Statements

- An if statement is a construct that enables a program to specify alternative path of execution.
- A one-way if statement executes an action if and only if the condition is true.
- The syntax for a one-way if statement is shown here:

```
if (boolean-expression)
{
    statement(s);
}
```

if Statements

```
if (radius >= 0)
{
    area = radius * radius * PI;
    cout << "The area for the circle of " <<
        " radius " << radius << " is " << area;
}
```



if Statements

- The boolean-expression is enclosed in parentheses:

```
if i > 0
{
    cout << "i is positive" << endl;
}
```

(a) Wrong

```
if (i > 0)
{
    cout << "i is positive" << endl;
}
```

(b) Correct

```
if (i > 0)
{
    cout << "i is positive" << endl;
}
```

(a)

Equivalent

```
if (i > 0)
    cout << "i is positive" << endl;
```

(b)

if Statements

- Check number status:

Enter an integer: 4
HiEven

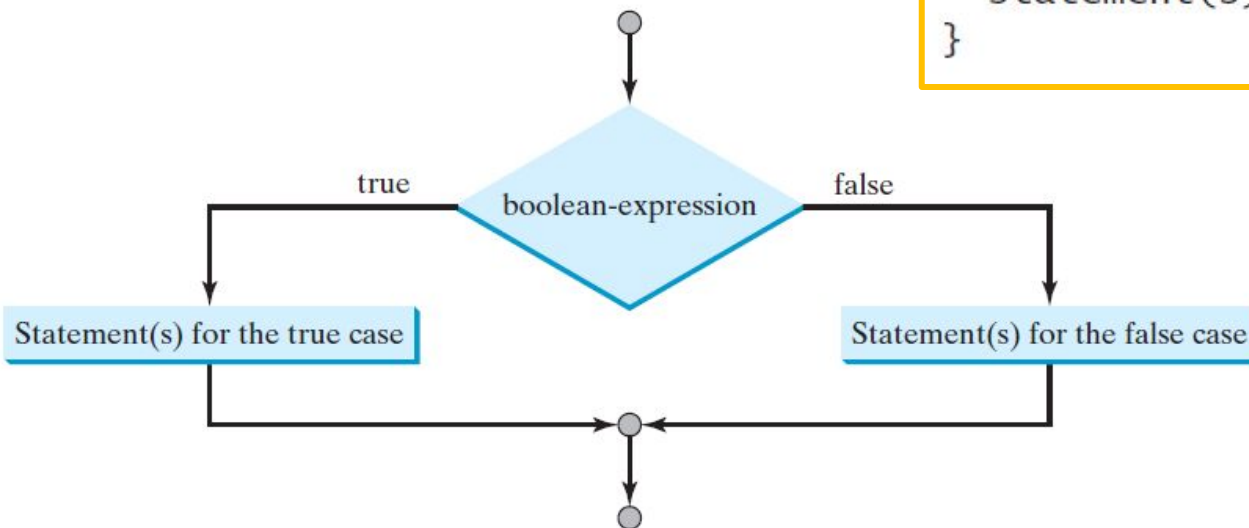
Enter an integer: 30
HiFive
HiEven

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      // Prompt the user to enter an integer
7      int number;
8      cout << "Enter an integer: ";
9      cin >> number;
10
11     if (number % 5 == 0)
12         cout << "HiFive" << endl;
13
14     if (number % 2 == 0)
15         cout << "HiEven" << endl;
16
17     return 0;
18 }
```


Two-Way if-else Statements

- An if-else statement decides which statements to execute based on whether the condition is true or false.

```
if (boolean-expression)
{
    statement(s)-for-the-true-case;
}
else
{
    statement(s)-for-the-false-case;
}
```



Nested if and Multi-Way if-else Statements

- An if statement can be inside another if statement to form a nested if statement.

```
if (i > k)
{
    if (j > k)
        cout << "i and j are greater than k" << endl;
}
else
    cout << "i is less than or equal to k" << endl;
```

Nested if and Multi-Way if-else Statements

- An if statement can be inside another if statement to form a nested if statement.

```
if (score >= 90.0)
    cout << "Grade is A";
else
    if (score >= 80.0)
        cout << "Grade is B";
    else
        if (score >= 70.0)
            cout << "Grade is C";
        else
            if (score >= 60.0)
                cout << "Grade is D";
            else
                cout << "Grade is F";
```

Equivalent

```
if (score >= 90.0)
    cout << "Grade is A";
else if (score >= 80.0)
    cout << "Grade is B";
else if (score >= 70.0)
    cout << "Grade is C";
else if (score >= 60.0)
    cout << "Grade is D";
else
    cout << "Grade is F";
```

This is better

Common Errors and Pitfalls

- Forgetting necessary braces, misplacing semicolons in an if statement, mistaking `==` for `=`, and dangling else clauses are common errors in selection statements.
- Duplicated statements in if-else statements and testing equality of double values are common pitfalls.

Common Errors and Pitfalls

- **Common Error 1: Forgetting Necessary Braces**
 - The braces can be omitted if the block contains a single statement.

```
if (radius >= 0)
    area = radius * radius * PI;
    cout << "The area "
        << " is " << area;
```

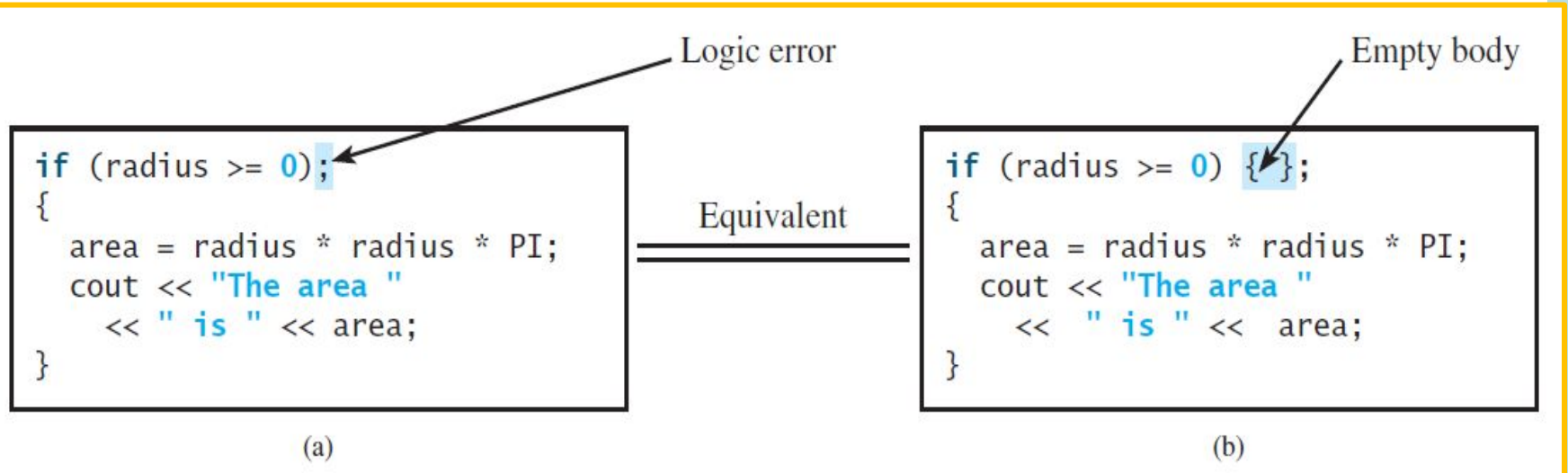
(a) Wrong

```
if (radius >= 0)
{
    area = radius * radius * PI;
    cout << "The area "
        << " is " << area;
}
```

(b) Correct

Common Errors and Pitfalls

- Common Error 2: Wrong Semicolon at the if Line



Common Errors and Pitfalls

Common Error 3: Redundant Testing of Boolean Values

- To test whether a bool variable is true or false in a test condition, it is redundant to use the equality testing operator like the code in (a):

```
if (even == true)  
    cout << "It is even.";
```

(a)

Equivalent

This is better

```
if (even)  
    cout << "It is even.";
```

(b)

Common Errors and Pitfalls

Common Error 4: Dangling else Ambiguity

```
int i = 1, j = 2, k = 3;  
  
if (i > j)  
    if (i > k)  
        cout << "A";  
else  
    cout << "B";
```

(a)

Equivalent

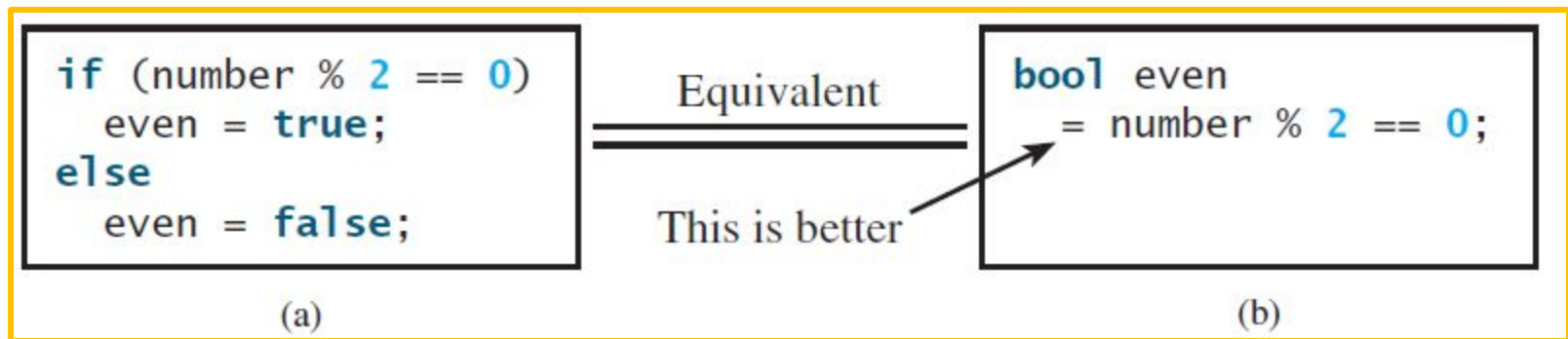
This is better
with correct
indentation

```
int i = 1, j = 2, k = 3;  
  
if (i > j)  
    if (i > k)  
        cout << "A";  
else  
    cout << "B";
```

(b)

Common Errors and Pitfalls

Common Pitfall 1: Simplifying Boolean Variable Assignment



Common Errors and Pitfalls

Common Pitfall 2: Avoiding Duplicate Code in Different Cases

```
if (inState)
{
    tuition = 5000;
}
else
```

```
{
    tuition = 15000;
}
cout << "The tuition is " << tuition << endl;
```

```
if (inState)
{
    tuition = 5000;
    cout << "The tuition is " << tuition << endl;
}
else
{
    tuition = 15000;
    cout << "The tuition is " << tuition << endl;
}
```

Logical Operators

- The logical operators `!`, `&&`, and `||` can be used to create a compound Boolean expression.

<i>Operator</i>	<i>Name</i>	<i>Description</i>
<code>!</code>	not	logical negation
<code>&&</code>	and	logical conjunction
<code> </code>	or	logical disjunction

Logical Operators

- Truth Table for Operator!

<i>p</i>	<i>!p</i>	<i>Example (assume age = 24, weight = 140)</i>
true	false	<code>!(age > 18)</code> is false, because <code>(age > 18)</code> is true.
false	true	<code>!(weight == 150)</code> is true, because <code>(weight == 150)</code> is false.

- Truth Table for Operator &&

<i>p1</i>	<i>p2</i>	<i>p1 && p2</i>	<i>Example (assume age = 24, weight = 140)</i>
false	false	false	<code>(age > 18) && (weight <= 140)</code> is true, because
false	true	false	<code>(age > 18)</code> and <code>(weight <= 140)</code> are both true.
true	false	false	<code>(age > 18) && (weight > 140)</code> is false, because
true	true	true	<code>(weight > 140)</code> is false.

Logical Operators

- Truth Table for Operator `||`

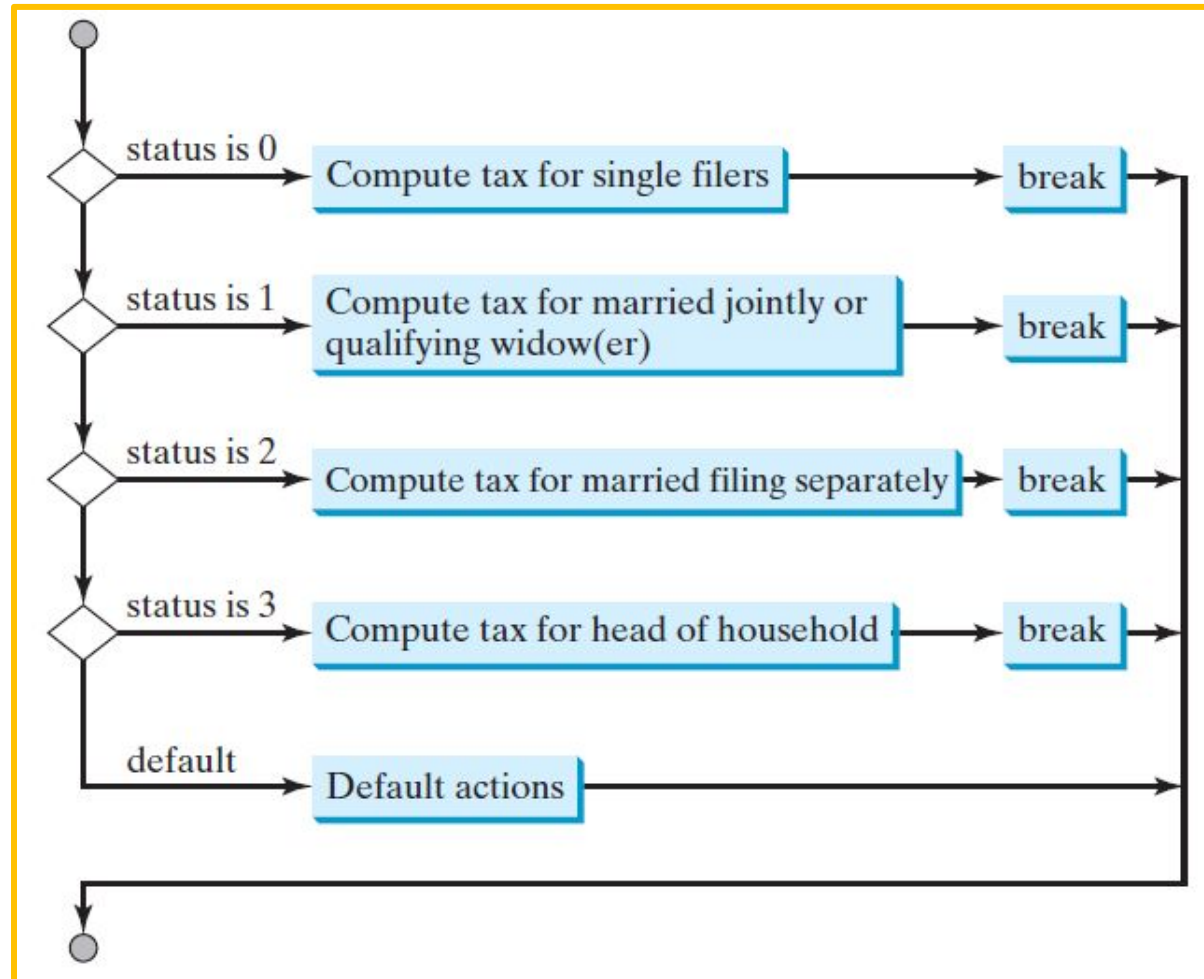
<i>p1</i>	<i>p2</i>	<i>p1 p2</i>	<i>Example (assume age = 24, weight = 140)</i>
false	false	false	<code>(age > 34) (weight <= 140)</code> is true , because <code>(weight <= 140)</code> is true .
false	true	true	
true	false	true	<code>(age > 34) (weight >= 150)</code> is false , because <code>(age > 34)</code> and <code>(weight >= 150)</code> are both false .
true	true	true	

switch Statements

- A switch statement executes statements based on the value of a variable or an expression.

```
switch (status)
{
    case 0:  compute tax for single filers;
             break;
    case 1:  compute tax for married jointly or qualifying widow(er);
             break;
    case 2:  compute tax for married filing separately;
             break;
    case 3:  compute tax for head of household;
             break;
    default: cout << "Error: invalid status" << endl;
}
```

switch Statements



switch Statements

```
switch (switch-expression)
{
    case value1: statement(s)1;
                break;
    case value2: statement(s)2;
                break;
    ...
    case valueN: statement(s)N;
                break;
    default:    statement(s)-for-default;
}
```


W5 - Practice

Exercises 1

Write a program in C++ to check whether the number is even or odd.

Expected Output:

```
Enter Number: 15  
15 is Odd number!
```

Exercises 2

Write a program in C++ to input a numeric grade and convert to a letter grade (if-else)

90 .. 100	A
80 .. 89	B
70 .. 79	C
60 .. 69	D
0 .. 59	F

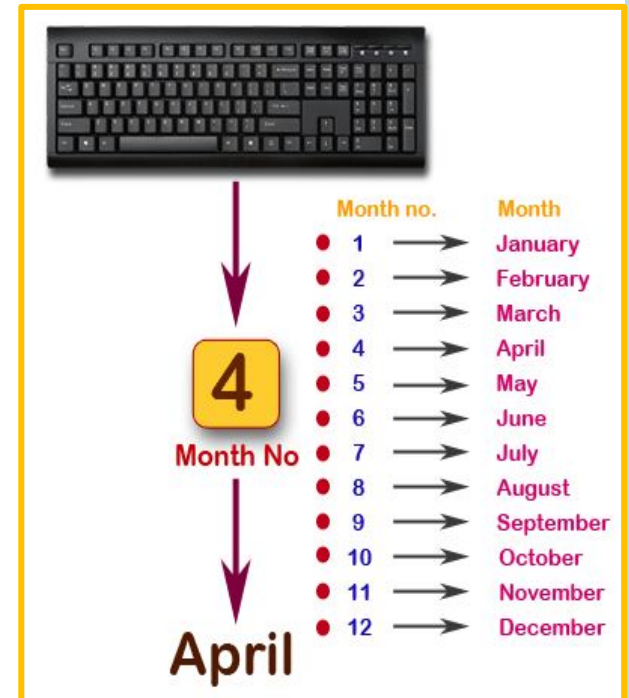
Expected Output:

```
Check the Grade:  
Enter score between 0-100: 85  
85 : B
```

Exercises 3

Write a program in C++ to read any Month Number in integer and display the Month name in the word.

Expected Output:



```
Please enter the month as a number (1-12): 4
April
```

Thanks!